Sara Maleubre Molinero

Cosmology – Large Scale Structure – N-Body Simulations – Dark Energy

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Education

2019-*Present* **PhD *Candidate* École Doctoral Physique en Île-de-France**, *Sorbonne Université - LPNHE*, Paris, France.

2017–2019 **MSc Astronomy and Cosmology**, *Leiden University*, Leiden, Netherlands.

2013–2017 **BSc in Physics**, *Universidad Autónoma de Madrid (UAM)*, Madrid, Spain.

2015–2016 **BSc in Physics (Erasmus Student)**, *Université Joseph Fourier*, Grenoble, France.

Research Experience

Oct-Mar 2022/23 **DAAD PhD Scholar**, Redshift-Space Distorsions in Euclid Flagship galaxy mock catalogue with split densities, Supervised by: Dr. Ariel Sanchez, *Max-Planck-Institute for Extraterrestrial Physics*.

2019-*Present* **PhD *Candidate* Thesis**, Theory of cosmological structure formation in the non-linear regime, Supervised by: Prof. Michael Joyce, *LPNHE, Sorbonne Université*.

2018-2019 **Second Master's Research Project**, Study of viable Horndeski models within the Planck phenomenological parametrization of Dark Energy, Supervised by: Dr. Alessandra Silvestri, *Lorentz Institute, Leiden University*.

2017–2018 **First Master's Research Project**, The density-environment of radio-selected galaxies, Supervised by: Dr. Jarle Brinchmann, *Leiden Sterrewacht, Leiden University*.

2016–2017 **Theoretical Final Bachelor Thesis**, Local effect of AGN on the host galaxy, Supervised by: Dr. Yago Ascasibar, *Universidad Autónoma de Madrid*.

2016–2017 **Experimental Final Bachelor Thesis**, Spectroscopic analysis of KIC 8462852, Supervised by: Dr. Carlos Eiroa, *Universidad Autónoma de Madrid*.

Summer 2016 **Summer Undergraduate Research (Erasmus Traineeship)**, High redshift galaxies and the reionization of the universe, Supervised by: Dr. Fergus Cullen, *The Institute for Astronomy, University of Edinburgh*.

Scholarships and Awards

Oct–Mar 2022/23 **DAAD Short-term Grant**, Salary, *Deutscher Akademischer Austauschdienst*.

2019–2022 **Jean-Pierre Aguilar Scholarship**, Salary + travel allowance, *Fondation CFM pour la Recherche*.

2018–2019 **Oort Scholarship**, Monthly Stipend, *Leiden University*.

Summer 2014, 2015, 2016 **Advanced Courses - Universidad Internacional Menendez Pelayo**, Full course grant, (Quantum physics and information, XIII International School of Astrobiology Josep Comas I Solá and General English courses), *Spanish Government*.

2013–2017 **Scholarships for Post-Obligatory studies, Undergraduate level**, Year Allowance + 2nd, 3rd and 4th year University tuition fee, *Spanish Government*.

2013 **High School Honourable mention**, 1st year University tuition fee, *Spanish Government*.

Selected Scientific and General Public Talks/Posters

October 2022 **Cosmology seminar at MPA**, (Germany)
The path to percent accuracy with Scale-Free Simulations

October 2022 **OPINAS seminar at MPE (Invited talk)**, (Germany)
The path to percent accuracy with Scale-Free Simulations

June 2022 **European astronomical society congress (Poster),** (Spain)

Accuracy of power spectra in dissipationless cosmological simulations

April 2022 **Congrès des Doctorants (Invited outreach talk)** ED-STEP'UP (France)

Successes and challenges of the Standard Cosmological model

April 2022 **CTC Cosmology Seminars (Invited talk)** Maryland Astronomy Center for Theory and Computation, University of Maryland (USA)

Accuracy of Power Spectrum from N-body simulations using Scale-Free cosmologies

March 2022 **Journal Club - CMB (John Kovac) group (Invited talk)** Harvard-Smithsonian Center for Astrophysics (USA)

Power Spectrum's accuracy in N-body simulations using Scale-Free cosmologies

March 2022 **Cosmology (Daniel Eisenstein's) group meeting (Invited talk)** Harvard-Smithsonian Center for Astrophysics (USA)

Scale-Free cosmologies to test Abacus accuracy

January 2022 **56th Rencontres de Moriond,** (Italy)

Accuracy of power spectrum measurements in Scale-Free simulations

October 2021 **Journal Club - Physical Cosmology Team (Invited Talk),** Ludwig-Maximilians-Universität München (Germany)

Power Spectrum's accuracy in N-body simulations using Scale-Free cosmologies

March 2021 **Journal Club - ICAP (Invited Talk),** Institute d'Astrophysique de Paris (France)

Power Spectrum measurements in Scale-Free Simulations

+2 posters, +6 scientific talk, +1 outreach talk

Teaching Experience/Mentoring/Organization

2022-Present **Co-organizer MPA/MPE -Cosmology- Journal Club**.

2022 **Organizer Early Career Journal Club - LPNHE**.

Fall 2021 **Experimental Physics**, (Year 3 - Minor Physics).

Fall 2021 **Optics and Electronics**, (Year 1 - Mathematics, Computer Sciences, Physics and Engineering).

Fall 2020 **Experimental Physics**, (Year 3 - Minor Physics).

2020-2022 **Mentor for Year 3 students with Master and Internship application**.

2018-2019 **Sterrenwacht Buddy Program (Mentoring)**, 3 international students.

Computer skills

Regular user Python, Mathematica

Occasional user git, SQL, C, C++, Julia

Publications

S. Maleubre, D. Eisenstein, L. H. Garrison, M. Joyce *Constraining accuracy of pairwise velocities using scale-free models.* 2022, *submitted* MNRAS

S. Maleubre, D. Eisenstein, L. H. Garrison, M. Joyce *Accuracy of power spectrum in dissipationless cosmological simulations.* 2022, MNRAS, 512, 1829

M. Leroy, L. Garrison, D. Eisenstein, M. Joyce, **S. Maleubre** *Testing dark matter halo properties using self-similarity.* 2021, MNRAS, 501, 5064

Additional Education

Music Degree, *The Spanish equivalent of Advanced Higher (A-levels) Music Qualification, with a specialization in* ***piano and composition***.